

5.2 - Further Classification of Operators

lemma 5.19 Let α be an operator on a finite-dimensional vector space V . Then,

(a) $\{0\} = \text{Ker}(\alpha^0) \subseteq \text{Ker}(\alpha) \subseteq \dots \subseteq \text{Ker}(\alpha^k) \subseteq \dots$

(b) if m is a non-negative integer such that $\text{Ker}(\alpha^m) = \text{Ker}(\alpha^{m+1})$, then $\text{Ker}(\alpha^m) = \text{Ker}(\alpha^{m+1}) = \text{Ker}(\alpha^{m+2}) = \dots$

lemma 5.20 Let α be an operator on an n -dimensional vector space V . Then, $\text{Ker}(\alpha^n) = \text{Ker}(\alpha^{n+1})$

lemma 5.21 Let α be an operator on an n -dimensional vector space V . Then, $V = \text{Ker}(\alpha^n) \oplus \text{Im}(\alpha^n)$

def 5.22 if α is a operator on a vector space V with eigenvalue λ , then a **generalized eigenvector** corresponding to λ is a vector $v \neq 0$ such that, for some positive int. j ,

$$(\alpha - \lambda \text{Id})^j(v) = 0$$

the **generalized eigenspace** corresponding to λ is the set $G_\lambda(\alpha)$ of all of the generalized eigenvectors corresponding to λ together w/ the zero vector

cor. 5.23 Let α be a linear operator on a vector space V of dimension n . Then, $G_\lambda(\alpha) = \text{Ker}(\alpha - \lambda \text{Id})^n$

lemma 5.24 if α is an operator and $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of α w/ generalized eigenvectors $v_1, \dots, v_m$ respectively, then the set $\{v_1, \dots, v_m\}$ is linearly independent

def 5.25 an operator α is **nilpotent** if, for some k , $\alpha^k = 0$

lemma 5.26 Let α be an operator on a vector space V and p any polynomial w/ coefficients in $\mathbb{F}$. Then $\text{Ker}(p(\alpha))$ and $\text{Im}(p(\alpha))$ are invariant under α

thm 5.27 | Generalized Eigenspace Decomposition

Let V be a finite dimensional complex vector space and α an operator on V . Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of α . Then,

(a) $V = G_{\lambda_1}(\alpha) \oplus \dots \oplus G_{\lambda_m}(\alpha)$

(b) each $G_{\lambda_j}(\alpha)$ is invariant under α

(c) When restricted to the subspace $G_{\lambda_j}(\alpha)$, the operator $\alpha - \lambda_j \text{Id}$ is nilpotent

Chapter 6 - Characteristic & Minimal Polynomials, Jordan Form, & the Determinant

def 6.1 | Let α be an operator on a vector space V , and λ an eigenvalue. The algebraic multiplicity of λ is $\dim(G_{\lambda}(\alpha))$. The geometric multiplicity is $\dim(E_{\lambda}(\alpha))$

def 6.2 | The characteristic polynomial of a linear map α with eigenvalues $\lambda_1, \dots, \lambda_m$ each with ^(algebraic) multiplicity $d_1, \dots, d_m$ is $\chi_{\alpha}(z) = (z - \lambda_1)^{d_1} \dots (z - \lambda_m)^{d_m}$

thm 6.3 | Cayley-Hamilton - for any complex vector space and operator α , the linear map $\chi_{\alpha}(\alpha)$ is the zero map

thm 6.4 | There is a unique polynomial $p \in \mathbb{C}[\lambda]$ of minimal degree with leading coefficient 1 such that $p(\alpha) = 0$

def 6.5 | The minimal polynomial of a linear map α is the monic polynomial $p(x)$ of smallest degree such that $p(\alpha)$ is the zero map

lemma 6.6 | Let α be a nilpotent operator. Then there are vectors $v_1, \dots, v_n \in V$ and non-negative integers $m_1, \dots, m_n$ such that $\{v_1, \alpha(v_1), \dots, \alpha^{m_1}(v_1), v_2, \alpha(v_2), \dots, \alpha^{m_2}(v_2), \dots, v_n, \dots, \alpha^{m_n}(v_n)\}$ is a basis for V , and $\alpha^{m_i+1}(v_i) = 0$ for all i

def 6.7 | A **Jordan basis** for an operator α is a basis with respect to which the matrix of α consists of blocks on the diagonal, and zeroes everywhere else, and the blocks all have the form

$$\begin{pmatrix} \lambda_j & 1 & & 0 \\ & \ddots & \ddots & \\ 0 & & \ddots & 1 \\ & & & \lambda_j \end{pmatrix} \quad \text{for some } \lambda_j$$

thm 6.8 | **Jordan Normal Form** - for every operator on a finite-dimensional complex vector space, there is a Jordan Basis of V for that operator

def 6.9 | The **trace** of a linear map is the sum of its eigenvalues, counted with multiplicity. The **determinant** is the product of the eigenvalues, counted with multiplicity